Sobol Sensitivity Analysis Assignment

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Assignment Part I

Choose one of the 3 papers below that provide an example of sensitivity analysis of model parameters. *Or* find another paper that uses model sensitivity analysis as part of understanding an envirnomental topic

After going through the paper, write a paragraph describing how results of the sensitivity analysis reported on in the paper might contribute to understanding (or prediction) within an environmental problem solving or management context.

As in, *why* did they do a sensitivity analysis? Why is it important here?

Selection: Uranium in Groundwater Model

Response: This study by Kumar et al. (2019) uses a Sobol sensitivity analysis to determine the effects of variables such as water intake rate (IR), uranium concentration (Cw), and exposure frequency (EF) on hazards of uranium. They used data from the Gaya District of Bihar, India, where they systematically sampled and tested groundwater sources. The study included two models: ingestion (oral) and dermal, which were conducted for child and adult age groups. Sensitivity analysis was used to determine the relative importance of each input. Results showed that Cw was the most sensitive input followed by IR and EF. It also found that adults experienced relatively higher exposure doses than the children in the region.

The sensitivity analysis allowed the researchers to predict which factors contributed most to oral and dermal uranium risk based on their model. In the future, they can compare this sensitivity analysis to real-world data with the same/similar variables, and determine how accurate and robust their model is (i.e. increase their understanding). If there model is accurate, it can then be applied and used in various regions to help inform regulation. Additionally, knowing the sensitivity of their model can help ensure that no one input (like IR) is overly sensitive; if this was the case, there would be a risk of large and inaccurate deviations in output based on small changes in input (i.e. sensitivity analysis can improve prediction).

Assignment Part 2

Recall our model of atmospheric conductance

$$C_{at} = \frac{v_m}{6.25 * ln(\frac{z_m - z_d}{z_0})^2}$$

$$z_d = k_d * h$$

$$z_0 = k_0 * h$$

$$z_m$$

is the height at which windspeed is measured - must be higher than the vegetation (cm), it is usually measured 200 cm above the vegetation

$$h$$

is vegetation height (cm)

$$v$$

is windspeed (cm/s)

Typical values if k_d and k_o are 0.7 and 0.1 respectively (so use those as defaults)

Your task

Repeat the sensitivity analysis that we have been working on in class BUT lets assume that we are in a different locations - where windpeeds are substantially higher and more variable AND vegetation is shorter - See details below

Consider the sensitivity of your estimate to uncertainty in the following parameters and inputs

$height$

k_d

k_0

v

Windspeeds v are normally distributed with a mean of 300 cm/s with a standard deviation of 50 cm/s

For vegetation height assume that height is somewhere between 3.5 and 5.5 m (but any value in that range is equally likely)

For the k_d and k_0 parameters you can assume that they are normally distributed with standard deviation of 1% of their default values

```r
# load necessary libraries
library(here)
library(sensitivity)
library(tidyverse)
```

Use the Sobel approach to generate parameter values for the 4 parameters:

```r
# source atmospheric conductance function
source(here("R/Catm.R"))

# decide on number of initial parameter sets
np <- 1000

# generate two examples of random number based on NEW assignment 5 parameter distributions
k_o <- rnorm(mean = 0.1, sd = 0.1 * 0.1, n = np)
k_d <- rnorm(mean = 0.7, sd = 0.7 * 0.1, n = np)
v <- rnorm(mean = 300, sd = 50, n = np)
height <- runif(min = 3.5, max = 5.5, n = np)

# create parameter set 1
x1 <- cbind.data.frame(k_o, k_d, v, height)

# repeat sampling (run variables again)
k_o <- rnorm(mean = 0.1, sd = 0.1 * 0.1, n = np)
k_d <- rnorm(mean = 0.7, sd = 0.7 * 0.1, n = np)
v <- rnorm(mean = 300, sd = 50, n = np)
height <- runif(min = 3.5, max = 5.5, n = np)

# create parameter set 2
x2 <- cbind.data.frame(k_o, k_d, v, height)
```

Run the atmospheric conductance model for these parameters:

```r
# create null sobol model using our two sets of parameters
sens_Catm_Sobol = sobolSalt(model = NULL, x1, x2, nboot = 100)

# access the different model parameters and assign proper column names
params <- as.data.frame(sens_Catm_Sobol$X)
colnames(params) <- colnames(x1)

# run the Catm model for all parameter sets and save results, using purrr package
```

```r
res <- pmap_dbl(params, Catm)

# indicate results to original Sobol object; this creates Sobol indices within model
sens_Catm_sobol = sensitivity::tell(sens_Catm_Sobol, res, res.names = "gs")
```

Estimate the Sobel Indices for output:

Show main effect, or S, indices. S indices partition variance (main effect without co-variance), and sum approximately to one.

```r
# assign row names based on corresponding parameters (for better understanding)
rownames(sens_Catm_sobol$S) = colnames(params)

# view main effect indices
sens_Catm_sobol$S
```

```
        original          bias std. error min. c.i. max. c.i.
k_o    0.2034946  0.0026737456 0.03276049 0.1384529 0.2579473
k_d    0.1994759  0.0036027109 0.03452185 0.1323609 0.2657004
v      0.4802659 -0.0011799143 0.02063420 0.4507592 0.5284301
height 0.1361409  0.0005130271 0.03189308 0.0620119 0.1914275
```

Show total effect, or T, indices, which account for parameter interactions.

```r
# assign row names based on corresponding parameters (for better understanding)
rownames(sens_Catm_sobol$T) = colnames(params)

# view total effect indices
sens_Catm_sobol$T
```

```
        original          bias std. error  min. c.i. max. c.i.
k_o    0.1966493  0.0009761201 0.01084423 0.17399389 0.2191402
k_d    0.1873027  0.0008759778 0.01164197 0.15821956 0.2106824
v      0.4891764 -0.0024875235 0.02504755 0.44534759 0.5385150
height 0.1140909 -0.0001529088 0.00801990 0.09705837 0.1294099
```

You can also view all indices by simply printing the model summary.

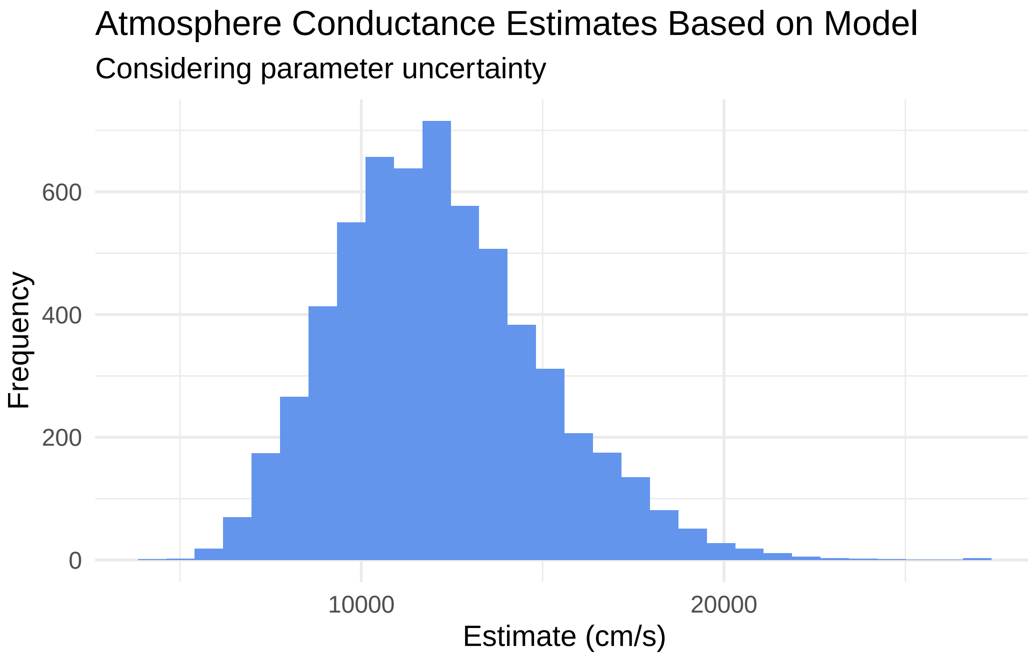
```r
print(sens_Catm_Sobol)
```

```
Call:
sobolSalt(model = NULL, X1 = x1, X2 = x2, nboot = 100)

Model runs: 6000

Model variance: 7073900

First order indices:
    original          bias std. error min. c.i. max. c.i.
X1 0.2034946  0.0026737456 0.03276049 0.1384529 0.2579473
X2 0.1994759  0.0036027109 0.03452185 0.1323609 0.2657004
X3 0.4802659 -0.0011799143 0.02063420 0.4507592 0.5284301
X4 0.1361409  0.0005130271 0.03189308 0.0620119 0.1914275

Total indices:
    original          bias std. error  min. c.i. max. c.i.
X1 0.1966493  0.0009761201 0.01084423 0.17399389 0.2191402
X2 0.1873027  0.0008759778 0.01164197 0.15821956 0.2106824
X3 0.4891764 -0.0024875235 0.02504755 0.44534759 0.5385150
X4 0.1140909 -0.0001529088 0.00801990 0.09705837 0.1294099
```

Plot conductance estimates in a way that accounts for parameter uncertainty:

```r
# first combine parameters and output
both <- cbind.data.frame(params, gs = sens_Catm_sobol$y)

# look at overall gs sensitivity to all parameter variation
ggplot(both, aes(x = gs)) +
  geom_histogram(fill = "cornflowerblue") +
  labs(x = "Estimate (cm/s)",
       y = "Frequency",
       title = "Atmosphere Conductance Estimates Based on Model",
       subtitle = "Considering parameter uncertainty") +
  theme_minimal()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

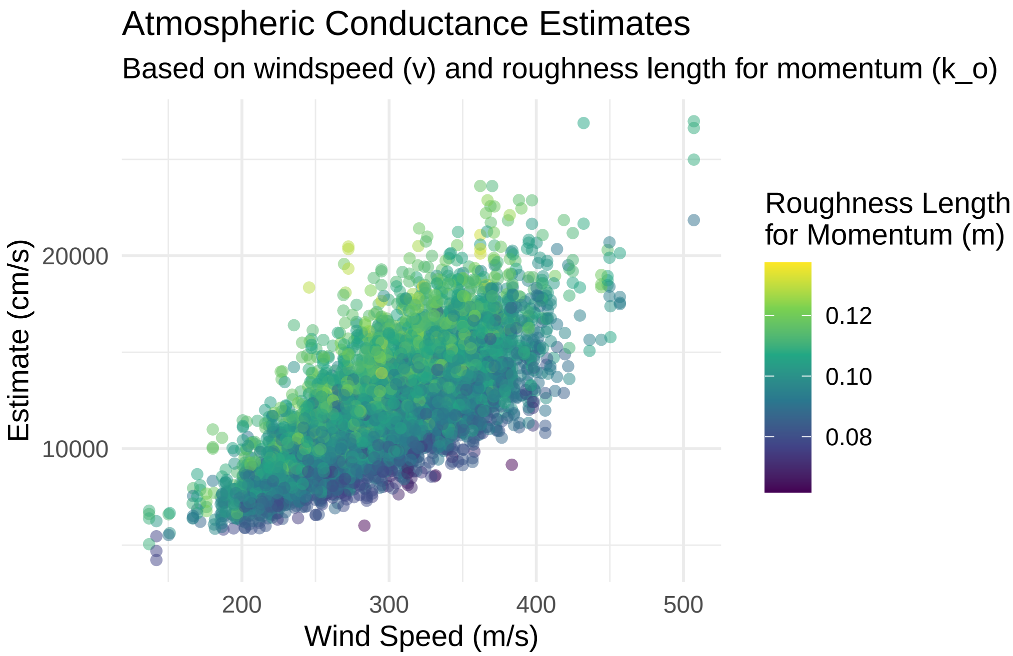


Plot conductance estimates against windspeed use the parameter that is 2nd in terms of total effect on response:

In my results, v (wind speed) has the highest *total* effect on response and k_d has the second. k_d and k_o are very close in effect (and results may change based on normal sampling of parameters), therefore I have included plots depicting both variables.

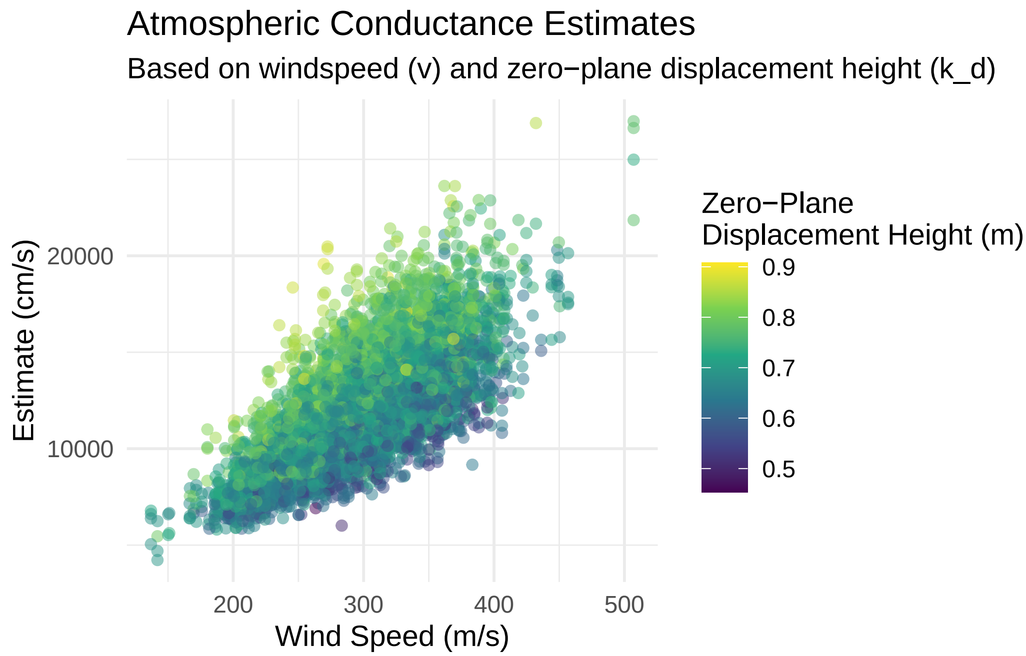
Conductance estimates against windspeed, using k_o as second variable:

```
# plot estimate relative to windspeed, coloring by k_d paremeter
ggplot(both, aes(x= v, y = gs, col = k_o)) +
  geom_point(alpha = 0.5) +
  labs(x = "Wind Speed (m/s)",
       y = "Estimate (cm/s)",
       title = "Atmospheric Conductance Estimates",
       subtitle = "Based on windspeed (v) and roughness length for momentum (k_o)",
       color = "Roughness Length \nfor Momentum (m)") +
  theme_minimal() +
  scale_color_viridis_c()
```



Conductance estimates against windspeed, using k_d as second variable:

```
# plot estimate relative to windspeed, coloring by k_d paremeter
ggplot(both, aes(x= v, y = gs, col = k_d)) +
  geom_point(alpha = 0.5) +
  labs(x = "Wind Speed (m/s)",
       y = "Estimate (cm/s)",
       title = "Atmospheric Conductance Estimates",
       subtitle = "Based on windspeed (v) and zero-plane displacement height (k_d)",
       color = "Zero-Plane \nDisplacement Height (m)") +
  theme_minimal() +
  scale_color_viridis_c()
```



Comment on what this tells you about how atmospheric conductance and its sensitivity to variation in windspeed differs in this setting as compared to the setting that we examined in class where windspeed was lower and less variable and vegetation was taller.

Wind speed has a much higher total effect on atmospheric conductance in *this* model/parameter set than the one we did in class. In the case of higher and more variable wind speeds as well as shorter vegetation, wind speed's total effect is ~0.5. In the previous example, it was ~0.2. This tells us that atmospheric conductance is more dependent on wind speed in the case of this assignment.

This suggests that the combination of higher wind speeds and shorter vegetation limit the effects of other variables in the system. It is difficult to exactly pinpoint whether it was vegetation height or wind speed that caused this effect, since we changed both at the same time. However, we can hypothesize:

- Shorter vegetation heights reduce the turbulence and mixing of air in canopies. Since this value decreased, it may have allowed for wind speed to have a larger effect.
- Higher wind speeds reduce the stagnant air boundary layer surrounding canopies. This may be very significant to atmospheric conductance, and therefore have a larger effect in this rendition of our model.

It is also interesting that total effects of k_o (roughness length for momentum, representative of surface drag) and k_d (zero_plane displacement height) change to be more equal with this parameter set. Overall, a change in parameters alters the effects of all model inputs.

Submit a pdf or html document with your code, graphs, and answers to the questions on Canvas Due

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Grading Rubric. Total points 80. (normalized to 10 for Canvas)

PART I (Paper Examples)

- Discussion of the implication of parameter uncertainty from example paper
  - explanation directly relates to a specific parameter (10pts)
  - explanation explores how parameter uncertainty might meaningfully impact results (10pts)

PART II (Atmospheric Conductance and Sobol)

- Generation of parameter values using Sobol (10pts)
- Running model for the parameters (10pts)
- Graph of uncertainty of the response variable
  - meaningful graph (5pts)
  - graphing style (axis labels, legibility) (5 pts)
- Graph of relationship between output and windspeed
  - choice of color (see instructions) (5pts)
  - graphing style (axis labels, legibility) (5 pts)
- Computing Sobol Indicators (10 pts)
- Discussion
  - correctly identifying how sensitivity to windspeed changed with setting (10pts)